

Sustainable Energy Planning Based on the Electrical Grid and Green Energy Transition in Kenya between 2019-2030

David Osigwar Okomol
Department of Energy and Built
Environment, Mount Kenya
University
Thika, Kenya
dokomol523@gmail.com

George Adwek
Department of Energy and Built
Environment, Mount Kenya
University
Thika, Kenya
adwekg@gmail.com

Joshua K. Ngoret
Department of Mechanical
Engineering, Jomo Kenyatta
University of Agriculture and
Technology- Kenya
jngoret@jkuat.ac.ke

Moses Arowo
Department of Chemical
Engineering, Moi University
Eldoret, Kenya
musarowo@yahoo.com

Abstract—This research article is a strategic and well holistic document to guide both national and county governments to achieve 100% energy access and prioritize sustainable energy transition besides creating many green jobs and enterprises. It shall also form a very primary document and basis for every investor across the various opportunities and, in particular, energy generation and distribution. Energy is the prime engine of economic growth; its planning signals a direct positive impact on social welfare, economic activities, socio-political development, environmental sustainability. This article uses Low Emissions Analysis Platform (LEAP) software to provide the respective Medium-Term Plan (MTP) & Long-Term Plan (LTP) Update for the period 2019 (base year) to 2030. This MTP is the identification and analysis of suitable expansion paths of the country's power system for that period, complying with the defined planning criteria and framework. This encompasses (i) examining of yesteryear electricity demand and advancement of future insistence scenarios; (ii) analysis of favorable expansion candidate fuels and technologies plus their scheduling; (iii) modeling of their anticipated addition to the prospective power generation and the expedient operation of the generation system for the year 2030. The key recommendation is that the Government of Kenya ought to determine means to sustainably expedite broad-based development in the MTP - implying stable and sufficient energy (including but not limited to grid electricity) supply to cater to growing demand.

Keywords—energy planning, demand forecasting, supply forecasting, green energy, Kenya

I. INTRODUCTION

The Government of Kenya ought to determine means to sustainably expedite broad-based development in the MTP - implying stable and sufficient energy (including but not limited to grid electricity) supply to cater to its ever increasing demand. The Kenyan Government is industriously seeking means to spur growth through improved private as well as public investment, with a specific target of dealing with the country's framework bottlenecks [1]. Energy planning & management is typified by

liberalization, deregulation, increased competition on various energy markets, and sustainability aspects [2]. The global aim as per SDG 7 - circumscribes three fundamental objectives: ensure cost-effective, sustainable, reliable, affordable and universal connection to up-to-date energy services; improve the percentage of renewable energy (RE) across the globe; and further double the global standard of energy efficiency enhancement [3]. The SDG 7 objectives contribute to the attainment of other SDG goals as illustrated in a number of studies [4, 5]. Regarding this, energy system planning is an undertaking of designating the resources and technologies required for energy generation, transmission and distribution, to meet community requirements [6]. Future energy demand is dependent upon the availability of natural resources, depletion potential, by which the dire need cater for environmental challenges [7-9]. New decision-support frameworks, methodologies, systematic and detailed planning frameworks are required to predict, analyse & determine future energy consumption. Sustainable energy planning in our context entails interconnected frameworks for the generation, transmission, distribution, and storage of different energy carriers such as gas, electricity, biomass, et cetera [6]. The integrated systems supply consumers with various kinds of end-use services & products; for instance, households require electricity for; ironing, heating, cooking using various appliances based on gas, electricity, oil, wood, etc. Distinctive kinds of energy consumption are industrial, office buildings, and commercial may have diversified consumption patterns [10]. A successful energy planning system can be adequately achieved through having a survey of the system's parameters, understanding all interactions involved in energy space, flows within and between the system as well as its surrounding. Physical boundaries typically correspond to the region's, locality's, town's or community's geographical borders within which energy flows having various system aspects that are measurable & allows for the system's performances such as reliability & efficiency to be determined [11]. In accordance with economic theories, energy

is a crucial resource for all industrial processes, whereby predicting energy consumption is a very critical phase for the industry's macro-planning & energy sectors [12]. Since it is complex to store some energy forms (e.g., electricity), it is vital to distribute electricity at least cost and minimal wastage. Adept planning for electricity distribution requires meticulous forecasts of future energy requirements to attain equilibrium for the supply & demand [13]. Forecasting glitches results to supply-demand imbalance, which negatively affects the network safety, operational cost, and service quality [14].

Energy consumption underestimation results in frequent supply interruptions due to outages, which negatively affects both economy and the society's daily operations. Besides, energy demand overestimation results to wastage of resources (mostly financial) due to unused capacity. Nevertheless, applying various models to precisely conduct future consumption patterns – especially non-linear data – is an indispensable affair for energy production & distribution systems [15]. Power system's base operations, e.g., unit maintenance, economic dispatch, fuel scheduling & unit commitment, are conducted more efficiently with accurate forecasts [16]. Forecasting investigations are also crucial for pricing. Energy price variations, rising energy demand, climate change, fossil fuels' over-reliance & under-developed new energies' affect energy security across the globe [17]. Environmental challenges, e.g., global warming, GHG emissions, are also crucial parameters of consumption forecasting. It was highlighted that > 75% of the anthropogenic GHGs resulted from fossil fuels burning. Fossil fuels usage is a typical means for electricity generation [18]. Hence, electricity generation approaches result in natural resources depletion and environmental pollution. Energy management has been imperative in the past due to the high rate of resource depletion, increased GHG emissions, and advancements in employing renewable, efficient, & clean energies are somewhat implemented. Demand forecasting is an important area in demand management [19]; various research has been performed on forecasting, indicating that the field is getting more attention.

Integrated energy planning considers key specifications with the potentials to vary over time, for instance, the macro framework e.g., inflation, economic advancement, global fuel costs et cetera; the population increase; technology changes; household; industrial level specific consumptions, and expected connectivity rates. To model and accurately forecast the energy consumption patterns for the near future, by applying various scenarios, for instance, economical & environmentally efficient development, GHGs reduction, social equity, responsible natural resources utilization et cetera [15]. Energy planning is fundamental in ensuring the system has adequate power generation capacity and a diverse mix of sources for more reliability, and also for optimal deployment depending on the prevailing demand situation. As a key step in mapping the available energy resources as well as strategizing on providing clean and sustainable energy to drive the Country's Integrated Development Plan (CIDP). The demand and supply will be analyzed in a bottom-up access to provide a reliable foundation for an energy demand forecast over the next 10 years.

II. METHODOLOGY

The assignment involved gathering data on the availability and utilization of RE and non-RE resources associated technologies, energy efficiency, and conservation measures with the aim to develop Kenya's sustainable energy plan. A bottom-up perspective is applied for the calculation of demand for domestic consumers for instance cooking, street lighting, and flagship projects as identified in Vision 2030. Sensitivity is conducted using three different scenarios; Business as Usual (BAU) scenario, enhanced scenario, and ambitious scenario.

A. BAU Scenario

Is the base case scenario with development projected from the historical growth.

B. Intermediate Scenario

The Intermediate strategy scenario was informed by the need to increase the uptake of RE within the country where the country seeks to reduce biomass wood dependency. The CEP document seeks to achieve a 25% reduction of non-renewable energies of households using wood fuel cooking technologies by 2025 and 100% by 2030.

C. Enhanced Scenario

This is an efficient and sustainable scenario. It's high electricity, solar, and LPG demand-driven scenario that advocates for a switch to clean biogas fuels, LPG, solar, and electricity. It is based on the patterns of the development and its highly driven by Vision 2030 growth extrapolations and the implementation of flagship strategies.

D. Ambitious Scenario:

This is a very ambitious RE use in urban centers. It looks to achieve zero biomass use in the country and high adaption of LPG, solar, and electricity. Quantitative data collected from the communities was analyzed using LEAP. The development of these indicators was based solely on the data collected throughout the study.

III. ENERGY DEMAND AND FORECASTING

LEAP adopts a demand-driven approach with a basic tree structure that defines consumption modules, transformation, and supply. The generic structure that will be adopted in the formulation of the energy plan. This will follow a bottom-up viewpoint for households and a top-down perspective for other sections to create a dependable base for an energy demand forecast that will feature ten years. The demand assessment covers the electrical power demand in isolated grids plus electrical power demand in unconnected areas, focusing on mechanical power and heat demands. The LEAP tool uses the demographic and microeconomics data in activity branches and GDP growth to simulate demand and projects a futuristic view of government actions [20, 21].

A. The reference year of account

The reference year provide data and information on the current year or the year with the most available data to inform realistic simulation. It should balance national energy consumption and supply in the current account. In most

occasions, census reports, recent surveys and actual import and export data are sort for. An energy balance is required to conform to the reference year national accounts. In this model, the census reports (2019), was used as the reference for the country demography. Focusing mainly on the use of RE, where electricity dominates the consumption, the model assumed the shares for electricity within different sectoral branches while factoring other forms of energies.

This provided detailed shares of all energy types demanded by households, Private Sector within top 100 consumers (this includes offices, private farms, private manufactures, etc.), Public sector within top 100 consumers (government institutions/offices, government agriculture farms, public schools, rural & urban development, SME Sector, Street Lighting, and Other Public and Private sector consumers. The energy demand distribution within Households, Private Sector within top 100 consumers, Public sector within top 100 consumers, SME Sector, Street Lighting, and Other Public and Private sector consumers were however deduced by expert judgment give the unavailability of data and limited time to collect such data as shown in Figure 1.

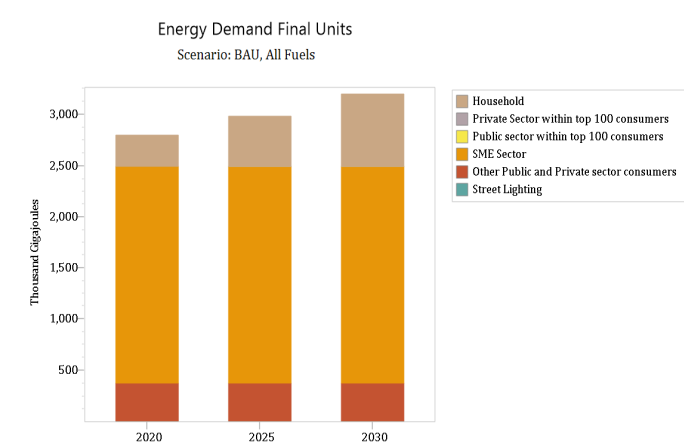


Fig. 1. The BAU Energy Demand

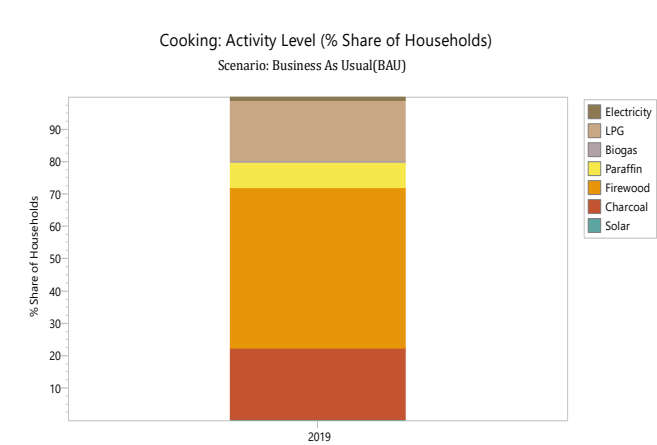


Fig. 2. The BAU Percentage of Households

To build the model and distribute technology adoption in the country, data was obtained from the Kenya energy sector fact sheet that was compiled from the KNBS, SCDIC, various ministries (national government) and other government agencies. Whereas the census reported energy consumption for cooking and lighting, didn't breakdown the technology use, we thus distributed total households using various wood and charcoal technology devices in the region with share fraction in Figure 2.

This representation provides an opportunity for target actions and tracking progress in adoption of specific technology. The intermediate strategy can similarly target a technology by targeting the population using the technology. It equally provides for a framework and a baseline for monitoring and evaluation of implementation of government actions. The model thus focuses on technology specific adoption and replacement for targeted actions as illustrated in Figure 3 and Figure 4.

Whereas wood fuel is the highly used energy source accounting for 71.8% overall demand in 2019, wood requirement for charcoal production also increases wood demand. This is due to highly inefficient conversion technology.

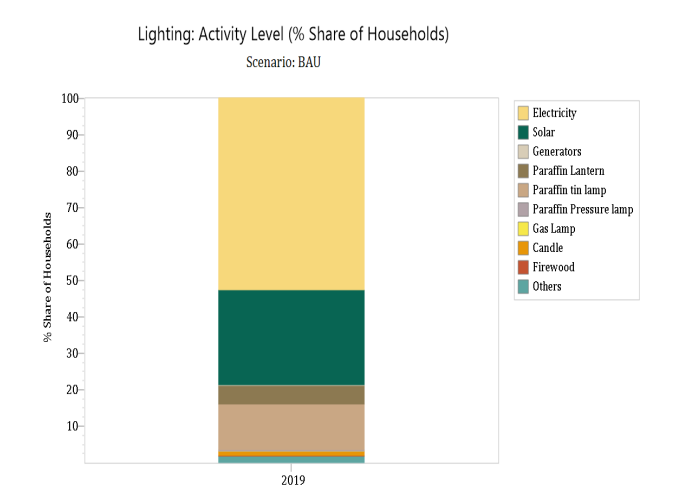


Fig. 3. Percentage Share of Lighting

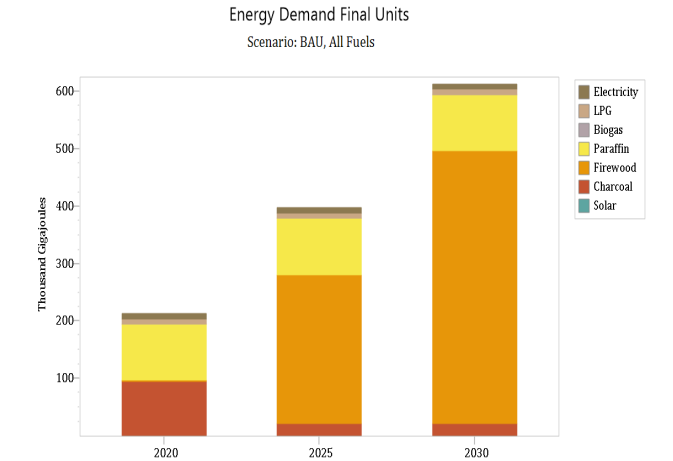


Fig. 4. BAU energy demand (all fuels)

B. Intermediate Scenario

The three A's of products penetration and adoptions (Availability, Affordability and Accessibility) were considered based on geospatial analysis while framing the possible targets per alternative technology. LPG is also an alternative to wood fuel given its penetration within rural and urban setting within the country. This scenario largely promotes the use of LPG and efficient charcoal conversion technologies. The highlights within this scenario are;

- 75% shift from wood fuel to LPG by 2030.
- 65% shift from other lighting technologies to electricity, with a 26% adoption of solar.
- There is a reduced dependency on wood and high promotion alternative fuels such as LPG and biogas
- There is the replacement of inefficient charcoal stoves in both urban and rural areas with LPG
- There is increased electrification and a significant rise in the use of LPG in urban areas.
- There is 75% penetration of LPG on average in all built areas by 2030.

The scenario if implemented will achieve a reduction of wood fuel use to about 25% in 2030, Figure 5. Achieving this however will require a concerted effort ranging from heightened awareness creation, compliance to regulations and appropriate policies.

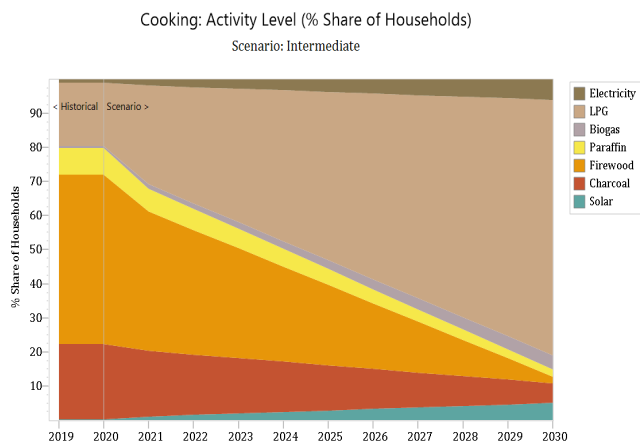


Fig. 5. Energy consumption trend

C. Enhanced Scenario

The actions that will help advocate for this are; promote fuel switching to green, affordable, and sustainable fuels as well as the cooking technologies through financial instruments, market transformation activities and social marketing/behavioural change communication strategies, capacity building, reforms of policy, subsidies, and relevant energy standards. This scenario if adopted will achieve the following; (i) the use LPG, biogas and, solar for cooking and heating, and the number of households using electricity will rise to 14% by 2030; (ii) there will be a

reduction in overall wood fuel dependency from 72% to around 3%. This scenario further develops the intermediate strategy, which calls for rapid promotion of alternative fuel uses such as electricity, solar, and LPG, Figure 5.

It assumes: (a) there is 6% use of non-renewable energy left in the country; (b) here will be a rise in electricity adoption to 14% from 0.9%, and 60% adoption of solar for lighting and heating; (c) there is 85% and 10% uptake of electricity and solar respectively in the top 100 consumers in the private sector; (d) A 43% uptake of electricity by the public sector from 15%, with solar and LPG taking majority of the remaining percentage. The SME sector will adapt electricity at 60% from 29.5% and a 10% use of solar energy. Other public and private sector consumers will adopt 45% electricity with 35% use of solar. Since this scenario looks to reduce the use of wood fuel to 0% by 2030, there will be high adoption of LPG to up to 15%.

D. Ambitious Scenario

The scenario advocates the use of only the most efficient devices in households and institutions.

The main assumptions in this scenario are:

- All households will use only electricity, solar, LPG, and biogas.
- The biogas adoption shall have increased to 5% in 2030, and there will be 55% adoption of LPG.
- About 10% of the total households shall be using electricity, and the remaining 30% will be solar by 2030.
- 50% of private sectors will take up the use of solar and 40% covered by electricity. There will be an 8% uptake of biogas and a 2% use of LPG with the sector.
- The public sectors' energy will be covered by 53% solar, 40% electricity, and distribution of the remaining energy supply within biogas and LPG.
- This scenario also tries to achieve a 50% penetration of solar energy in the SMEs and a 30% uptake of electricity. Given the nature of SMEs in the country being sparsely distributed, there will be minimal use of biogas at 9% and 10% uptake of LPG.
- Other public and private sectors will take up a 52% use of electricity and 40% solar application.
- There will be a shift of street lighting from electricity to solar that will take up 80%, 90% of household lighting will be taken up by solar, with the remaining 10% covered by electricity.

The actions within this strategy include; (i) development and harmonization of strategies and market chances for national biofuel industry growth such as biogas; (ii) capacity building and policy reforms can attract investors and the private sector to increase the sale of LPG and decentralized electrification as well as promote the adoption of solar energy.

IV. CONCLUSION AND RECOMMENDATIONS

This research sought to develop a sustainable energy planning models based on the electrical grid and green energy transition for the period 2019-2030 in line with the CIDP; the economic outlook for the country; committed power generation and transmission projects; grid extension and densification projects and updated energy demand for households, institutions and SMEs. Specific Challenges to delivering improved energy services under the CIDP and Vision 2030 MTP and beyond include: (i) mobility of populations, (ii) lack of technical know-how in installation / construction of various systems, (iii) high costs of installation & maintenance of RE systems, (iv) inadequate financing, technical skills to install & maintain RE technologies among communities, (v) security of installation - theft & vandalism of PV and wind energy systems, (vi) national government (Ministry of Energy and Petroleum (MoE&P) and REREC) project implementation timelines, (vii) environmental degradation due to increased use of biomass, (viii) lack of supportive energy policies, (ix) limited knowledge and awareness in relation to technology or energy supply options

From this research, the following conclusions and recommendations are provided;

Given the abundant wind & solar resources across the county, these resources provide immense opportunities within the country for a wide range of applications as follows;

The distributed nature of settlements in the country's population demands that lighting service should be provided using portable lighting devices like solar lanterns. The provision of lighting services should be implemented through a collaboration with private sector businesses/ SMEs. Specific recommendations with respect to solar, wood fuel, and wind include:

- a. Develop new mini-grids in the short term targeting areas with high economic potential such as central, coastal and western regions given that these installations can be applied when the locations are eventually interconnected to the national grid.
- b. Use PV and wind systems to supplement isolated grids given the resources are available throughout the year
- c. Encourage mobile RE solutions targeting pastoralist communities in the Eastern and North Eastern Kenya
- d. Mainstream technical training of technicians through TVET centers to support installation and maintenance of RE systems
- e. Community awareness creation & sensitization drive to increase community ownership of projects so as to enhance sustainability and minimize the cases of vandalism
- f. Address land issues to facilitate investment by the private sector in large scale Energy projects

- g. Liaison with REREC and MoE&P to accelerate projects critical to the county. For instance, synchronize initiative with Lake Turkana Wind Power (LTWP), MoE&P, and REREC

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